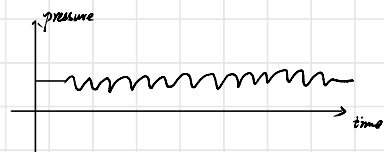
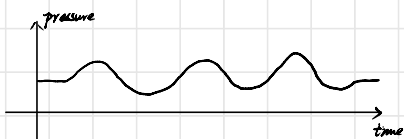


Intuition example: decomposing sound



it is easy to
add things up,
but how to
decompose
a complex signal
into sum of simple
signals?

fourier
transform



If we plot this figure: 0.5 cycles per second (1 cycle corresponding to 360°)

(In other words,

2 seconds $\rightarrow$ 1 cycle (360°)

6 beats

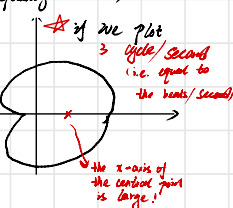
angle: 36° corresponding to 1 second in the

time-pressure figure

length = pressure

i.e. the y-axis

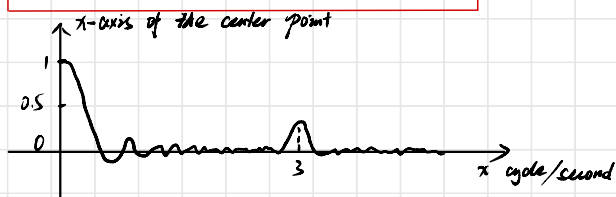
of the time-pressure
figure

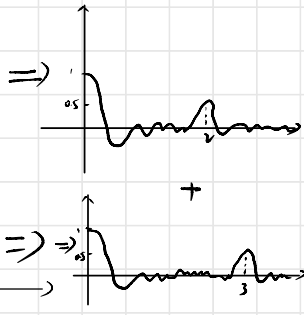
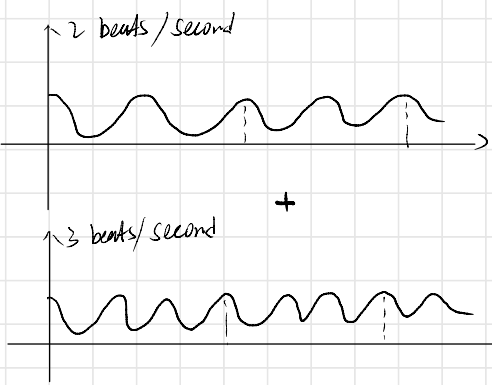
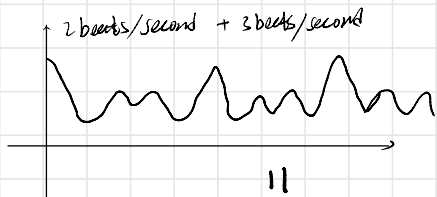


the x-axis of
the center point
is large.



The point: when the cycle/second = beats/second,
the x-axis of the center point is large.

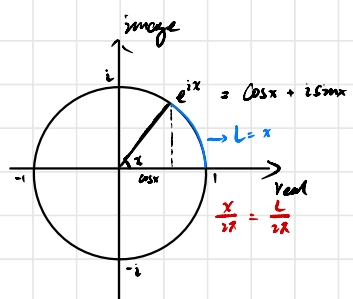




time domain

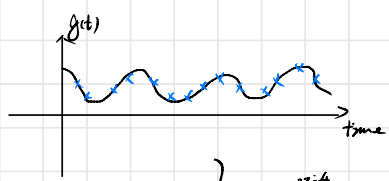
frequency domain

To describe winding and rotation, we use complex plane



Consider: e^{ixt}
 $\rightarrow$ rotate one cycle per second
 $e^{ix/2t}$
 $\rightarrow$ rotate one cycle per $1/2$ second
 $e^{-ix/4t}$
 $\rightarrow$ rotate in *clockwise* direction (as we did in last page)

$g(t)e^{-ixft}$
 $\rightarrow$ this controls the magnitude (length) of the rotation

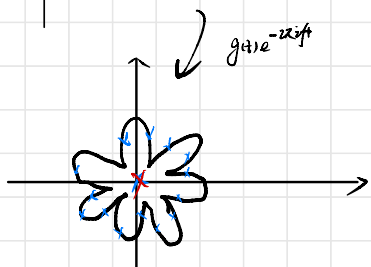


Estimate the center point by sampling from $g(t)$

$$\frac{1}{N} \sum_{k=1}^N g(t_k) e^{-ixft_k}$$

$\downarrow$ take limit

$$\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} g(t) e^{-ixft} dt$$



$\rightarrow$ *inverse* Fourier transform: multiply the center of mass by its time
 effect: when a frequency period for a long time it gets longer